OPTICAL MUSIC RECOGNITION FOR JAZZ LEAD SHEETS: A CRNN-CTC APPROACH FOR MONOPHONIC LINES

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GEP Deliverable 3: Budget and Sustainability

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Contents

1 Project Budget

This section presents the full economic budget for the project, following the structure recommended in the GEP economic management module. The budget is broken down into: (1) human resources costs, itemised per Gantt activity; (2) general costs (hardware, software, and overheads); (3) a contingency reserve; and (4) specific incidentals linked to risks identified in Deliverables 1 and 2.

1.1 Human Resources Costs

The project is executed by a single developer who assumes all professional roles. To estimate labour costs realistically, each task is assigned to the role that best describes the work performed, with an hourly rate derived from average gross salaries in Spain for the IT sector. Rates are grounded in the 2023 INE *Encuesta Anual de Estructura Salarial* (EAES) [1], the official Spanish salary survey published by the Instituto Nacional de Estadística. The national mean gross salary was €28,049.94 in 2023; workers in the ICT sector (Section J of NACE rev. 2) earn substantially above the national average, making the €30,000–38,000 gross range assumed here consistent with EAES data for junior-to-mid level ICT roles. These figures are adjusted by a ×1.35 multiplier to convert gross salary to total employer cost, which accounts for employer social security contributions [2], as prescribed in the GEP economic management module [3].

The four roles and their fully loaded hourly rates are:

Table 1: Role definitions and hourly rates (gross salary × 1.35 social security [2]).

Role	**Gross salary (€/year)**	**Loaded (€/year)**	**€/h**[1]
Project Manager	38 000	51 300	29.15
Research Analyst	33 000	44 550	25.31
ML Engineer	36 000	48 600	27.61
Technical Writer	30 000	40 500	23.01

Table 2 details the human resources cost for every task in the Gantt chart (Deliverable 2). Each task is assigned to its primary role; the cost is computed as hours × rate.

1.2 General Costs

General costs cover hardware, software, and infrastructure expenses that are shared across all project activities and cannot be attributed to a single task.

1.2.1 Hardware

Amortisation is calculated linearly: $\text{cost} = \text{price} \times \frac{\text{project months}}{\text{useful life} \times 12}$.

1.2.2 Software

All software used in this project is free/open-source or available through university licences at no additional cost:

Table 2: Human resources budget by Gantt task.

Code	Task	Role	h	€/h	Cost (€)
WP0 – Project Management (130 h)					
T0.1	Context and scope definition	Project Mgr.	20	29.15	583.00
T0.2	Temporal planning	Project Mgr.	12	29.15	349.80
T0.3	Budget and sustainability	Project Mgr.	15	29.15	437.25
T0.4	Final GEP integration	Project Mgr.	8	29.15	233.20
T0.5	Supervisor meetings	Project Mgr.	25	29.15	728.75
T0.6	Thesis writing	Tech. Writer	50	23.01	1 150.50
WP1 – Sprint 1: Literature Review & Baseline (78 h)					
T1.1	OMR literature survey	Research Analyst	25	25.31	632.75
T1.2	Dataset exploration	Research Analyst	15	25.31	379.65
T1.3	Classical CV baseline impl.	ML Engineer	25	27.61	690.25
T1.4	Baseline evaluation	ML Engineer	10	27.61	276.10
T1.5	Sprint 1 review	Project Mgr.	3	29.15	87.45
WP2 – Sprint 2: Dataset Construction (73 h)					
T2.1	Output encoding design	Research Analyst	12	25.31	303.72
T2.2	Synthetic rendering pipeline	ML Engineer	35	27.61	966.35
T2.3	Data augmentation	ML Engineer	15	27.61	414.15
T2.4	Validation and splits	ML Engineer	8	27.61	220.88
T2.5	Sprint 2 review	Project Mgr.	3	29.15	87.45
WP3 – Sprint 3: CRNN-CTC Development (75 h)					
T3.1	CRNN architecture design	ML Engineer	25	27.61	690.25
T3.2	Training pipeline impl.	ML Engineer	20	27.61	552.20
T3.3	Initial training run	ML Engineer	15	27.61	414.15
T3.4	Evaluation framework	ML Engineer	12	27.61	331.32
T3.5	Sprint 3 review	Project Mgr.	3	29.15	87.45
WP4 – Sprint 4: Iterative Refinement (68 h)					
T4.1	Error analysis	Research Analyst	10	25.31	253.10
T4.2	Data pipeline improvements	ML Engineer	15	27.61	414.15
T4.3	Model refinement runs	ML Engineer	25	27.61	690.25
T4.4	Ablation studies	ML Engineer	15	27.61	414.15
T4.5	Sprint 4 review	Project Mgr.	3	29.15	87.45
WP5 – Sprint 5: Final Evaluation & Defence (85 h)					
T5.1	Final evaluation	ML Engineer	15	27.61	414.15
T5.2	Qualitative error analysis	Research Analyst	10	25.31	253.10
T5.3	Extension exploration (cond.)	ML Engineer	20	27.61	552.20
T5.4	Final thesis writing	Tech. Writer	30	23.01	690.30
T5.5	Defence preparation	Tech. Writer	10	23.01	230.10
	TOTAL HUMAN RESOURCES		**509**		**13 640.77**

Table 3: Hardware costs with amortisation.

Item	Price (€)	Life (yr)	Use (mo)	Amort. (€)
Laptop (development)	1 200	5	5	100.00
GPU – NVIDIA RTX 3060 12 GB	350	4	5	36.46
Monitor (external)	250	6	5	17.36
Peripherals (keyboard, mouse)	80	4	5	8.33
Total hardware amortisation				**162.15**

Table 4: Software costs.

Software	**Licence**	**Cost (€)**
Python 3.14, PyTorch, OpenCV, music21	Open-source	0.00
LilyPond + LilyJAZZ	Open-source (GPL)	0.00
LaTeX (TeX Live)	Open-source	0.00
Git / GitHub	Free (education)	0.00
Zotero	Open-source	0.00
Linux (Fedora)	Open-source	0.00
Total software		**0.00**

Table 5: Infrastructure and overhead costs (5-month project duration).

Item	**Monthly (€)**	**Justification**	**Total (€)**
Electricity	15.00	Desktop PC + GPU $\approx$ 300 W $\times$ 5 h/day	75.00
Internet	10.00	Proportional share of home broadband	50.00
Workspace (home)	0.00	No additional rent (home office)	0.00
Cloud GPU (reserve)	–	Only if R4 materialises (see §1.4)	0.00
Total overheads			**125.00**

Table 6: Summary of general costs.

Category	**Cost (€)**
Hardware amortisation	162.15
Software	0.00
Overheads	125.00
Total general costs	**287.15**

1.2.3 Other Overheads

1.2.4 General Costs Summary

1.3 Contingency

A contingency reserve of **15%** is applied to the sum of human resources and general costs. This percentage is within the 10–20% range recommended for IT projects and reflects the moderate level of detail in this budget (task-level granularity, but single-person execution with inherent estimation uncertainty).

$$\text{Contingency} = 0.15 \times (13\,640.77 + 287.15) = 0.15 \times 13\,927.92 = \textbf{€2 089.19}$$

1.4 Incidentals

Incidental costs are derived from the risks identified in Deliverables 1 and 2 (Table 4 of Deliverable 2). Each incidental is costed at its full remediation cost and then weighted by the probability of occurrence.

[1] Based on 1 760 effective working hours per year (220 days × 8 h/day).

Table 7: Incidentals budget linked to project risks.

Risk	Description	Remediation action	Full cost (€)	Prob.	Weighted (€)
R1	Limited ground-truth data	Fall back to PrIMuS typeset images; 10 h extra augmentation work.	276.10	25%	69.03
R3	Model underfitting / CTC misalignment	Simplify vocabulary and retrain; 10 h extra ML Engineer time from T5.3 buffer.	276.10	20%	55.22
R4	Computational constraints	Rent cloud GPU (e.g., Lambda or Vast.ai) for 40 h at ∼€1/h + 5 h setup.	178.05	15%	26.71
R5	Scope creep	Reduce T5.3 to feasibility discussion (2–3 h writing); no extra hardware cost, only re-planned hours already budgeted.	0.00	30%	0.00
Total incidentals					**150.96**

Note on R2 (Staff detection errors): This risk affects only the optional extension task T5.3 and does not generate additional costs beyond re-allocation of time already budgeted.
Note on R5 (Scope creep): Although likely, the contingency action consists of re-scoping rather than spending additional resources, so its monetary impact is zero.

1.5 Total Budget

Table 8: Total project budget.

Budget item	Cost (€)
Human resources (Table 2)	13 640.77
General costs (Table 6)	287.15
Contingency – 15% (§1.3)	2 089.19
Incidentals (Table 7)	150.96
TOTAL PROJECT BUDGET	**16 168.07**

The total estimated project cost is **16 168.07 €**. Human resources represent 84.4% of the total, which is expected for a research-intensive software project with minimal hardware requirements. The contingency and incidentals together account for 2 240.15 € (13.9%), providing a reasonable safety margin.

2 Budget Management Control

Effective budget control requires both procedures and quantitative indicators to detect and correct deviations during project execution.

2.1 Control Procedures

1. **Biweekly time tracking:** Actual hours spent on each task are logged in a spreadsheet at the end of each biweekly cycle, coinciding with supervisor meetings. This provides the raw data for deviation analysis.

2. **Sprint-end cost review:** At the end of each sprint (every ∼4 weeks), the accumulated cost for the completed work package is compared against the planned budget. Deviations are documented and, if significant, trigger a re-forecast for remaining sprints.

3. **Risk register update:** At each sprint review, the risk register (Deliverable 2, Table 4) is revisited. If a risk materialises, the corresponding incidental budget is activated; if a risk is retired, its budget allocation is released to the contingency pool.

4. **Scope change protocol:** Any addition or removal of tasks requires explicit documentation and a budget impact assessment before approval.

2.2 Deviation Indicators

Three key performance indicators (KPIs) are tracked:

Cost Variance (CV)
: CV = Budgeted Cost of Work Performed (BCWP)−Actual Cost of Work Performed (ACWP). A positive CV indicates under-budget execution; a negative CV signals overspending. Action threshold: $|CV| > 10\%$ of the sprint budget triggers a corrective plan.

Schedule Variance (SV)
: SV = BCWP − Budgeted Cost of Work Scheduled (BCWS). Negative SV indicates schedule delay, which often leads to cost overruns. Action threshold: $SV < -5\%$ of sprint budget.

Cost Performance Index (CPI)
: CPI = BCWP/ACWP. A $CPI < 0.90$ at any sprint boundary triggers a detailed root-cause analysis and a revision of the estimate at completion (EAC).

These indicators follow Earned Value Management (EVM) principles [4] and are computed at each sprint review. If the CPI drops below 0.90 for two consecutive sprints, the contingency reserve (§1.3) is partially allocated to cover the overrun, and the scope of T5.3 (conditional extension) is reduced or eliminated to compensate.

3 Sustainability Report – Initial Milestone

This section addresses the initial-milestone sustainability questions (row "I" of the sustainability matrix defined in the GEP sustainability guide [5]; see also the economic management module [3]). The analysis covers the three mandatory dimensions: economic, environmental, and social.

3.1 Self-Assessment of Sustainability Competency

After completing the sustainability self-assessment survey, the following conclusions emerge:

- **Strengths:** I have a solid awareness of the economic dimension of projects, having studied project management and cost estimation during the degree. I understand the importance of quantifying resource consumption and can identify direct and indirect costs with reasonable confidence. My computing background also gives me an understanding of energy consumption in computational workloads.
- **Weaknesses:** My knowledge of environmental impact measurement beyond energy consumption (e.g., supply chain, e-waste, carbon offset mechanisms) is limited. On the social dimension, I tend to focus on the direct user benefits and underestimate the broader societal implications, both positive and negative. This project has prompted me to consider these aspects more carefully.
- **Growth areas:** I aim to improve my ability to reason about the full lifecycle impact of software systems, including deployment, maintenance, and eventual decommissioning. The sustainability matrix framework is a useful tool that I intend to apply beyond this thesis.

3.2 Economic Dimension

PPP – Project cost estimation. The total estimated cost of the project is 16 168.07 € (see §1.5). Human resources dominate the budget (84.4%), reflecting the research and development nature of the work. Material costs are low thanks to the exclusive use of open-source software and existing consumer hardware. The budget includes a 15% contingency reserve and risk-weighted incidentals, providing a realistic safety margin.

Useful life – Comparison with the state of the art. Currently, musicians and archivists who need to digitise jazz lead sheets must either transcribe scores manually (slow and expensive, typically €15–30 per page for a professional copyist) or use commercial OMR tools such as Audiveris, SmartScore, or PhotoScore, which range from free (with limited accuracy) to €100–250 for a licence. However, none of these tools are specifically tuned for the jazz lead-sheet idiom (handwritten-style fonts, chord symbols, specific clefs), resulting in high error rates that still require significant manual correction.

This project aims to produce an open-source, specialised model that, once trained, can be run at negligible marginal cost: inference on a single score image takes under one second on consumer hardware. If the model achieves a low SER, it would substantially reduce the cost of digitising lead-sheet collections compared to both manual transcription and generic commercial tools. The model weights, code, and training pipeline will be released publicly, eliminating licence costs for downstream users.

Useful life – Economic viability. The project itself is an academic exercise and does not generate direct revenue. However, its outputs (trained model, dataset, code) have potential economic value if integrated into open-source music tools (e.g., MuseScore, OpenScore) or adopted by music libraries and archives. The low operational cost (CPU-only inference, no cloud dependency) makes deployment economically viable even for small organisations with limited IT budgets.

3.3 Environmental Dimension

PPP – Environmental impact estimation. The main environmental impact of the project is electrical energy consumption for computation and personal work. We estimate:

- **Development workstation:** The laptop and monitor consume approximately 80 W during regular development (coding, writing, literature review). Over 509 project hours, this amounts to $509 \times 0.08\,\text{kW} = 40.7\,\text{kWh}$.
- **GPU training:** The NVIDIA RTX 3060 draws $\sim$170 W under training load. Estimated GPU-active time across all training tasks (T3.3, T4.3, T4.4, T5.1, T5.3): $\sim$95 h, yielding $95 \times 0.17\,\text{kW} = 16.2\,\text{kWh}$.
- **Dataset generation:** LilyPond rendering of $\sim$43 000 images is CPU-bound; estimated at $\sim$10 h of sustained CPU ($\sim$65 W), contributing 0.65 kWh.
- **Total:** $\sim$57.5 kWh over five months, equivalent to approximately 13.2 kg CO_2 at the 2023 Spanish grid emission factor of 0.23 kg CO_2/kWh [6], [7].

To put this in context, 57.5 kWh is roughly equivalent to running a household refrigerator for 4–5 days or driving a petrol car for approximately 55 km. The environmental footprint is modest.

PPP – Minimisation measures. Several measures are taken to reduce the environmental impact:

- **Efficient hardware:** The RTX 3060 was chosen for its strong performance-per-watt ratio, and the ResNet18 backbone is deliberately lightweight (vs. larger architectures like ResNet50 or Vision Transformers).
- **Mixed-precision training:** PyTorch AMP (Automatic Mixed Precision) reduces GPU memory usage and speeds up training, lowering total energy consumption per training run.
- **Early stopping:** Training halts automatically when validation loss plateaus, avoiding unnecessary computation.
- **Re-use of resources:** No new hardware was purchased for this project; all equipment was already owned and in use.

Useful life – Comparison with the state of the art. Manual transcription has negligible direct energy cost but consumes significant human time. Commercial OMR tools run on standard PCs with similar energy profiles. Cloud-based OMR services (e.g., Google Cloud Vision for document OCR) involve data-centre energy consumption that is typically higher per query than local inference. This project's model runs locally on consumer hardware, avoiding data-centre overhead. Furthermore, by releasing a trained model, the environmental cost of training is incurred *once* and amortised across all users, rather than each institution training its own model.

3.4 Social Dimension

PPP – Personal impact. Undertaking this project has been personally enriching in several ways:

- It has deepened my expertise in deep learning, computer vision, and music information retrieval – areas I am passionate about and intend to pursue professionally.
- It has strengthened my ability to plan and execute a medium-scale research project independently, including literature review, experimental design, and scientific writing.
- As a musician, working on OMR has made me more aware of the accessibility challenges faced by visually impaired musicians and the importance of machine-readable music formats for assistive technologies.

Useful life – Comparison with the state of the art and real need. Jazz lead sheets – particularly those from The Real Book – are among the most widely used musical documents in jazz education and performance worldwide. Despite their ubiquity, they remain largely in paper or scanned-image form, limiting searchability, transposability, and integration with digital music tools.

There is a genuine need for an accurate, open, and accessible OMR solution tailored to this repertoire:

- **Musicians** benefit from digital scores that can be transposed instantly for different instruments, annotated electronically, and shared via tablets during rehearsals and gigs.
- **Music educators** can use digitised lead sheets to create interactive learning materials, extract melodic fragments for exercises, and build searchable repertoire databases.
- **Music archivists and libraries** gain the ability to preserve and index large collections of jazz scores in machine-readable formats, improving long-term accessibility.
- **MIR researchers** obtain structured symbolic data for computational musicology, jazz analysis, and recommendation systems.

No group is expected to be adversely affected by the project. The system processes publicly available or user-owned scores and does not collect personal data. The open-source licence ensures that no vendor lock-in or dependency is created.

Useful life – Is there a real need? Yes. Existing OMR tools achieve reasonable results on typeset classical scores but perform poorly on the specific visual style of jazz lead sheets (handwritten-style fonts, chord symbols above the staff, informal layout). Musicians currently resort to manual transcription or accept low-quality automatic results. A specialised, accurate, and free tool directly addresses this unmet need.

References

[1] Instituto Nacional de Estadística (INE). "Encuesta anual de estructura salarial (EAES). año 2023. datos definitivos." Official Spanish annual earnings survey (INEbase). The national mean gross annual salary in 2023 was 28,049.94 EUR. ICT-sector workers (NACE Section J) earn substantially above the mean, consistent with the 30,000–38,000 EUR gross range used here., Accessed: Mar. 11, 2026. [Online]. Available: https://www.ine.es/dyngs/INEbase/es/operacion.htm?c=Estadistica_C&cid=1254736177025&menu=ultiDatos&idp=1254735976596

[2] Ministerio de Inclusión, Seguridad Social y Migraciones. "Cotización y recaudación de trabajadores. regímenes." Official employer social security contribution rates for the Spanish General Regime. Total employer overhead is approximately 30 percent, commonly rounded to a x1.35 multiplier on gross salary in academic project budgets, as prescribed by the GEP economic management module., Accessed: Mar. 11, 2026. [Online]. Available: https://www.seg-social.es/wps/portal/wss/internet/Trabajadores/CotizacionRecaudacionTrabajadores/10721

[3] Departament de Gestió, FIB–UPC, *Project management – module 2.4: Economic management*, GEP course material (internal). Covers budget structure (staff costs, general costs, amortisation, contingency, incidentals) and prescribes the x1.35 salary multiplier for Spanish employer social security contributions., Facultat d'Informàtica de Barcelona (FIB), Universitat Politècnica de Catalunya, 2025.

[4] Project Management Institute, *A Guide to the Project Management Body of Knowledge (PMBOK Guide)*, 7th ed. Newtown Square, PA, USA: Project Management Institute, 2021, ISBN: 978-1-62825-664-2.

[5] Departament de Gestió, FIB–UPC, *Sustainability analysis for the bachelor's thesis*, GEP course material (internal). Defines the sustainability matrix (Environmental, Economic, Social x PPP, Exploitation, Risks) and the questions for the initial and final milestones., Facultat d'Informàtica de Barcelona (FIB), Universitat Politècnica de Catalunya, 2018.

[6] Red Eléctrica de España (REE) / Redeia, "Informe del sistema eléctrico español 2023," Red Eléctrica de España, Tech. Rep., 2024, Annual report of the Spanish electricity system. The CO2 emission intensity of the Spanish grid in 2023 was approximately 0.23 kg CO2/kWh, as also reported by MITECO. Accessed: Mar. 11, 2026. [Online]. Available: https://www.ree.es/es/datos/publicaciones/informe-anual-sistema

[7] Ministerio para la Transición Ecológica y el Reto Demográfico (MITECO). "Factores de emisión: Electricidad." MITECO publishes annual CO2 emission factors for the Spanish electricity mix. The 2023 factor is approximately 0.23 kg CO2/kWh., Accessed: Mar. 11, 2026. [Online]. Available: https://www.miteco.gob.es/es/cambio-climatico/temas/mitigacion-politicas-y-medidas/factores_emision_tcm30-479095.pdf